

Week 8, October 30, 2024

Main topics

- Generalized Tikhonov and Total Variation.
- Iterative solutions of 2D problems

Material

From last week: P. C. Hansen, *Discrete Inverse Problems: Insight and Algorithms*, Chapter 8.
New material this week: No new material.

Research lecture on Tuesday October 29 by Professor Michael Vogelius, Rutgers University - New Brunswick; currently visiting Aarhus University in 303b/134.

Title: Passive manipulation of electromagnetic fields.

Abstract: In this talk I shall discuss some theoretical results concerning invisibility, non-scattering, and second harmonic generation. The results are in frequency domain and are formulated in the context of the Helmholtz equation.

Program

13:00-13:30 Generalized Tikhonov, Total Variation and sparsity

13:30-17:00 Exercises, see below

Exercises

1. In the lecture, we introduce the matrix L , which represent the first order derivative operator, based on differences between neighbouring elements of the solution vector x , representing samples of the solution at the grid points $t_j = (j - \frac{1}{2})/n$, $j = 1, \dots, n$. Alternatively, we can use central differences that involve elements with doubled distance, i.e., x_{j-1} and x_{j+1} . Show that the corresponding matrix for the first derivative are given by

$$\hat{L}_1 = \frac{1}{2} \begin{pmatrix} -1 & 0 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & -1 & 0 & 1 \end{pmatrix}.$$

Also show that the null spaces of this matrix is

$$\mathcal{N}(\hat{L}_1) = \text{span}\{(1, 1, \dots, 1)^T, (1, -1, 1, -1, \dots)^T\}.$$

Why is it delicate to use $\hat{L}_1$ in the regularization term?

2. This exercise illustrates that Tikhonov regularization, corresponding to $L = I$, is not always the best choice in regularization problems. We use the Abel problem to make the point.

A simple way to compute the Tikhonov solutions is to use the least squares formulation

$$x_\lambda = \arg \min_x \left\| \begin{pmatrix} A \\ \alpha L \end{pmatrix} x - \begin{pmatrix} y \\ 0 \end{pmatrix} \right\|_2^2.$$

In MATLAB you simply write as usual

$$x = [A; \alpha * L] \setminus [y; \text{zeros}(p, 1)];$$

where p is the number of rows of L . Here are the steps of the exercise.

- Reconsider the Abel problem; Generate the matrix $A \in \mathbb{R}^{n \times n}$, $n = 128$. Fix the function $f(t) = t^2 - 2t \cos(2 * \pi * t)$, discretize on a uniform grid to a vector x and compute $y_{\text{exact}} = Ax$.
- Add Gaussian white noise to the exact right-hand side, with standard deviation 0.005.
- Compute ordinary Tikhonov solutions and tune the regularization parameter α until the noise starts to dominate x_α , and plot the solutions. Find the optimal solution (explain your criteria for optimality).
- Generate the first derivative smoothing matrix L_1 . Compute the generalized Tikhonov solutions with L_1 . Again, vary λ until the noise starts to dominate $x_{L, \lambda}$. Find and plot the optimal solution.
- Repeat the previous step with the second derivative matrix

$$\hat{L}_2 = \frac{1}{4} \begin{pmatrix} 1 & -2 & 1 & & \\ & \ddots & \ddots & \ddots & \\ & & 1 & -2 & 1 \end{pmatrix}.$$

- Compare the optimal solutions from the three different regularization terms. What are the main differences?
 - Play in your experiments with a regularization term $\alpha \|x - x_0\|^2$ for some choices of x_0 .
3. For the above three choices of L (i.e., I , L_1 and L_2) use the L-curve to determine the regularization parameter. It is ok to “eyeball” this value from a rough plot of the L-curve. Note that the three values of λ might be quite different.
4. Let f be the indicator function on the interval $(-1/2, 1/2)$. Compute $\text{TV}(f)$ using the definition in weak form.

Consider the function in $L^2((-1, 1)^2)$ given by

$$f(x) = \begin{cases} a & |x| < 1/2, \\ 0 & |x| \geq 1/2. \end{cases}$$

Compute $\text{TV}(f)$ via the definition and the divergence theorem.

- Download the 1D TV regularization solver `TVreg.m` from “Additional Matlab software” on <http://people.compute.dtu.dk/pcha/Regutools>. (For python/julia users, find another implementation.) For the same problem as in exercise 2, use TV-regularization to compute a solution. Find the optimal parameter; compare the results.
- Repeat exercise 2 and 5 with the function

$$f(t) = \begin{cases} t & 0 \leq t \leq 1/4 \\ -1/2 & 1/2 \leq t \leq 2/3 \\ 0 & \text{otherwise.} \end{cases}$$

- The Sobolev norm is

$$\|x\|_{H^1(a,b)}^2 = \|x\|_{L^2(a,b)}^2 + \|x'\|_{L^2(a,b)}^2$$

and hence a discrete version of the corresponding regularization term is

$$\alpha\|x\|_2^2 + \beta\|L_1x\|_2^2$$

with the two regularization parameters $\alpha, \beta > 0$.

How would you numerically implement the problem

$$x_{\alpha,\beta} = \arg \min_x J_{\alpha,\beta}(x), \quad J_{\alpha,\beta}(x) = \|Ax - y\|_2^2 + \alpha\|x\|_2^2 + \beta\|L_1x\|_2^2?$$

Reconsider the problem from exercise 2. Solve the problem with this regularization. Optimize for the values of α and β .

- Continue from last week (exercise 7) the solution method for the 2D forward heat equation that given initial heat distribution $x(s) = x(s_1, s_2)$, produces

$$y(t_1, t_2) = Kx(t_1, t_2) = \int k(t, s)x(s) \, ds$$

with the kernel (now corrected with correct constant)

$$k(t, s) = \frac{4}{\pi} \sum_{n,m} e^{-(n^2+m^2)T} \sin(ns_1) \sin(ms_2) \sin(nt_1) \sin(mt_2).$$

- Write a function taking as input samples of x on a uniform grid and giving as output samples of y on the same grid. Use `fft` – do not try to set up the matrix.
- Test your code on

$$x(s_1, s_2) = \frac{2}{\pi} \sin(ns_1) \sin(ms_1);$$

what should the exact y be?